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An Optimized Graph-Theoretic Framework for Energy-Aware Virtual Network Mapping in Cloud-Based Medical Imaging Systems

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ABSTRACT: With the development of the Internet and the use of the Internet for medical information, the demand for large-scale and reliable management of medical information has led to large-scale Internet data centers. This work shows the initial formulation of a medical information model, which is developed to help medical workers and medical departments share information, integrate resources, and analyze information. This software consists of amount of medical information and gives services from medical data centers. Virtual Network Mapping (VNM) is about building a network based on demand by installing virtual machines in a substrate network, subject to limitations on capacity, bandwidth, and latency. It is significant to data centers for dealing with dynamic cloud workloads. Simultaneously, the mapping of virtual and physical networks with several providers consumes more time and energy. In order to resolve this issue, VNM has been mapped by making use of Graph Theory (GT) matching, a well-studied database topic. (i) Brain-Storm Optimization Graph Theory (BSOGT) is introduced for modelling a virtual network request in the form of a GT with different resource constraints, and a substrate network is treated as a graph, where the nodes and edges contain at-tributes that specify their constraints. (ii) The newly introduced algorithm uses graph decomposition into topology patterns, which is then followed by BSOGT to resolve the problem of mapping. (iii) In this model, ERVNM and its BSOGT are formulated with a mapping energy computation function. (iv) The problem is viewed as a virtual network mapping to be the ACGT, and integer linear programming is used to get the optimal solutions. The performance of the ACGT, heuristic, exact, and two-stage algorithms is evaluated by means of a cloud Simulator environment. The results obtained from the simulation indicate that the BSOGT algorithm attains the objectives of cloud service providers with respect to Convergence Time, Mapping percentage, Acceptance ratio, and Processing time.

KEYWORDS: Leaf disease, Virtual Network Mapping (VNM), Brain-Storm Optimization Graph Theory (BSOGT), Cloud Networking, Virtualization, Quality of Services (QoS), Distributed, Cloud Computing (CC), and Optimization.

1 Introduction

The success of Cloud computing in the form of a paradigm arises from the concepts that form the basis of its implementation. Rendering the client with the capability of stretching or shrinking the rented infrastructure in a dynamic manner on demand is a key benefit over the conventional plan and buy model. This feature, integrated with delivery over quick connections and the elaborate pay-as-you-go pricing choices, describes the popularity of the cloud amongst business clients. Nonetheless, the concept, which helped in the implementation of all these features, is virtualization. By means of virtualization, cloud service providers can provide computational power, storage, platforms, and services in a commodity-dependent design without clients having to bother about the implementation details at the low level. Clients can perform the computation and connect with no overhead of management of resources or network routing and control [1]. In order to accomplish this goal, cloud service providers require systems with efficiency to completely control the resource virtualization, allocation, and scheduling in cloud data centers. Cloud data centers house thousands of servers that deal with storing, processing, and exchanging the data of the clients.

Network virtualization has not been in the radar of focus of several researchers in recent times. It enables the process of making multiple virtual networks over one single physical network. However, in spite of this benefit, network virtualization is affected by the problem of mapping the virtual links and nodes to the physical network in the most effective manner. Meanwhile, management of both interdata center and intradata center networks is entirely carried out by the cloud provider. This issue about Virtual Network Mapping with resource and energy constraints is referred to as ERVNM. It is described as the problem of mapping the virtual nodes and links to the physical nodes and paths [2].

VNM is crucial in the management of big data. Big data is frequently distributed to the data centers [3]. Nonetheless, data center networks tend to become a drawback for dynamic cloud workloads, comprising querying and controlling the data. In conventional networking platforms, the configuration of network resources is done manually with static policies, and new workload provisioning mostly takes days or weeks. This provides a highlight on the necessity for VNM, for the automatic deployment of virtual networks in a data center network as a response to requests made in real-time. Rather, VNM is finding increasing use in industry, e.g., Amazon's EC2, VMware Data Center, and Big Switch Networks. It is efficient in maximizing the server usage and minimizing server provisioning time (reducing from days or weeks to minutes), server capital expenses, and operating expenditures. Also, there has been a lot of work carried out on virtualization methodologies for big data and database systems. Virtualization of network resources involves running several logical networks over the same physical infrastructure simultaneously. For creating a virtual network, mapping the virtual nodes and links on the physical infrastructure, also referred to as Routing and Wavelength Assignment RWA, is an important step. The virtualization can form an abstraction between the user and physical resources, where the user is provided with the illusion of directly interacting with the physical resources. Otherwise said, virtualization can conceal the specifications concerned with the network infrastructure [4].

VN embedding problem clients can also make connection requests. These connections will help in the data exchange to happen between the client VMs or between a client VM and the private cloud of that client. The VMs can denote the vertices of a virtual network where every client is expected to maintain the accepted Quality-of-Service (QoS) criteria with no regard to the fact about the number of other clients, who are sharing the data center resources simultaneously. A key criterion here is the capability of the system to assign the network resources to the VMs in a dynamic manner at any point in time. However, getting a mapping between the virtual machines and hosts provides a solution to a part of the problem of mapping a virtual environment to the physical environments. One more problem is getting, for every link between two virtual machines, a path in the physical infrastructure stretching from the host that runs one of the virtual machines to the host that runs the other one. The issue behind the virtual network embedding itself is an extreme challenge already, and is NP-hard even in the case of offline. Consequently, a variety of heuristic and optimization-based algorithms have been shown in the relevant literature. However, pre-selection of

node mappings with no consideration to their association with the link mapping phase confines the solution space, and it can lead to poor performance [5].

In this research, a better correlation is introduced between the virtual and physical resources by introducing the Brain-Storm Optimization Graph Theory (BSOGT). In the case of BSO, new energy value will be produced in order to get some of the old energy replaced if the BSO algorithm provides poor mapping results between the physical and virtual network mapping. Whether a BSOGT provides poor mapping results or not is decided by means of the length of the Hamiltonian cycle that can be got by having all the physical node constraints transferred into an undirected weighted graph. The virtual nodes are mapped onto substrate nodes in a manner that enables the mapping of virtual links to physical paths in the subsequent phase. The new algorithm utilizes the graph decomposition into topology patterns by employing BSOGT for solving the mapping problem. Providing both in data center and in-network hosting by considering network nodes to be able to offer determined and storage resources in addition to switching and routing functions. The newly introduced BSOGT algorithm was tested and assessed against the available algorithms, making use of comprehensive simulations that indicate that the newly introduced BSOGT algorithm performs better than the other algorithms.

2 Literature Review

Calheiros et al [6] suggested a heuristic for mapping both the virtual machines to hosts and the virtual links running between virtual machines to paths in the actual system. It is also essential to take into account that the Virtual Machine Monitor (VMM) exploits the resources of the host. Moreover, the problem addressed by the issue is defined, a solution is presented for it, and it is evaluated in a variety of usage conditions. Guo et al [7] examined the issue of shared backup network provisioning for VN embedding and advised on two shared backup network provision strategies for virtual network embedding, which include the Shared On-Demand approach (SOD_BK) and the Shared Proactive approach (SP_BK). By sharing the bandwidth utilized by the restoration flows from various VNs, the necessary backup bandwidth can be largely minimized, and a larger amount of substrate resources can be saved for accepting VN requests in the future. Experimental Simulation indicates that both the newly introduced mechanisms make the best usage of substrate resources compared to the dedicated backup mechanism with no sharing, whilst each one of them has its individual benefits.

Yeow et al [8] concentrated on the issue of resource allocation for Virtual Infrastructure (VInfs) embedding with the guarantee of reliability. As a physical infrastructure that hosts several VInfs, it offers more re-source efficiency in sharing redundant nodes between VInfs. The physical footprint cast by redundant links can be minimized by taking the maximum over every failure scenario when allocating the resources with a linear program taken from the Multi-Commodity Flow problem. Sun et al [9] designed a framework for the optimal provisioning of VN requests by making use of mixed integer programming. In addition, the efficiency of this technique is demonstrated in enhancing the overall revenue by carrying out elaborate simulations on various networks.

Abedifar et al [10] introduced a Particle Swarm Optimization (PSO) concept used for virtual network mapping. Five diverse cost functions are devised, and a novel encoding technique for optical networks is presented. The newly introduced optimization mechanism is simulated by getting the mapping of various virtual networks onto a physical infrastructure for optimizing five diverse cost functions. The results are provided and explained for specified cost parameters. Alhazmi et al [11] introduced an extensive system for solving virtual network mapping performed for a set of connection requests transmitted by cloud clients. Connections are gathered in time intervals known as windows. The results obtained from the simulation indicate that the dynamic window size algorithm accomplishes the objectives of the cloud service providers about the ratio of served connections, resource usage, and computational overhead.

Cao et al [12] introduced the modelling of a virtual network request in the form of a graph pattern with multiple constraints, and a substrate network is treated as a graph, where the nodes and edges have attributes indicating their capability. In addition, it is shown that several mapping needs can be expressed in this

model, like virtual machine placement, network embedding, and priority mapping. It is verified experimentally that these algorithms are effective and are capable of finding high-quality mappings, making use of real-life and artificial data. Alzahrani et al [13] studied about energy sensitive particle swarm optimization algorithm to be used for distributed clouds. This algorithm targets partitioning every Virtual Network Request ("VNR") to subgraphs, making use of the Heavy Clique Matching technique ("HCM") for generating a coarsened graph. Every coarsened node present in the coarsened graph is allocated to an appropriate Data Center ("DC"). The newly introduced algorithm was tested and assessed against the available algorithms, employing comprehensive simulations that indicate that the newly introduced algorithm performs better than the other algorithms.

Xiao et al [14] studied a Global Resource Capacity -Survivable Virtual Network Mapping embedding algorithm (IntD-GRC-SVNE), which provides the implementation of multidomain mapping in network virtualization. The results obtained from the simulation indicate that IntD-GRC-SVNE cannot just enhance the survivability of a multidomain communications network but also make the network load more balanced and hugely enhance the network acceptance rate owing to the usage of GRC (global resource capacity). Mechtri et al [15] provided the solution to the problem of cloud and networking resources mapping observed in distributed and hybrid cloud environments. A precise algorithm is introduced for mapping the joint nodes and links of the virtual infrastructure graph requested to the physical graph from many providers. In order to deal with the complexity and scalability for huge virtual and physical networks containing thousands of nodes, an effective heuristic algorithm, depending on topology patterns and bipartite matching, is utilized for providing near-optimal solutions and minimizing the mapping delays by an order of three to four magnitude.

3 Proposed Methodology

In this research, a better correlation is introduced between the virtual and physical resources by presenting the Brain-Storm Optimization Graph Theory (BSOGT).

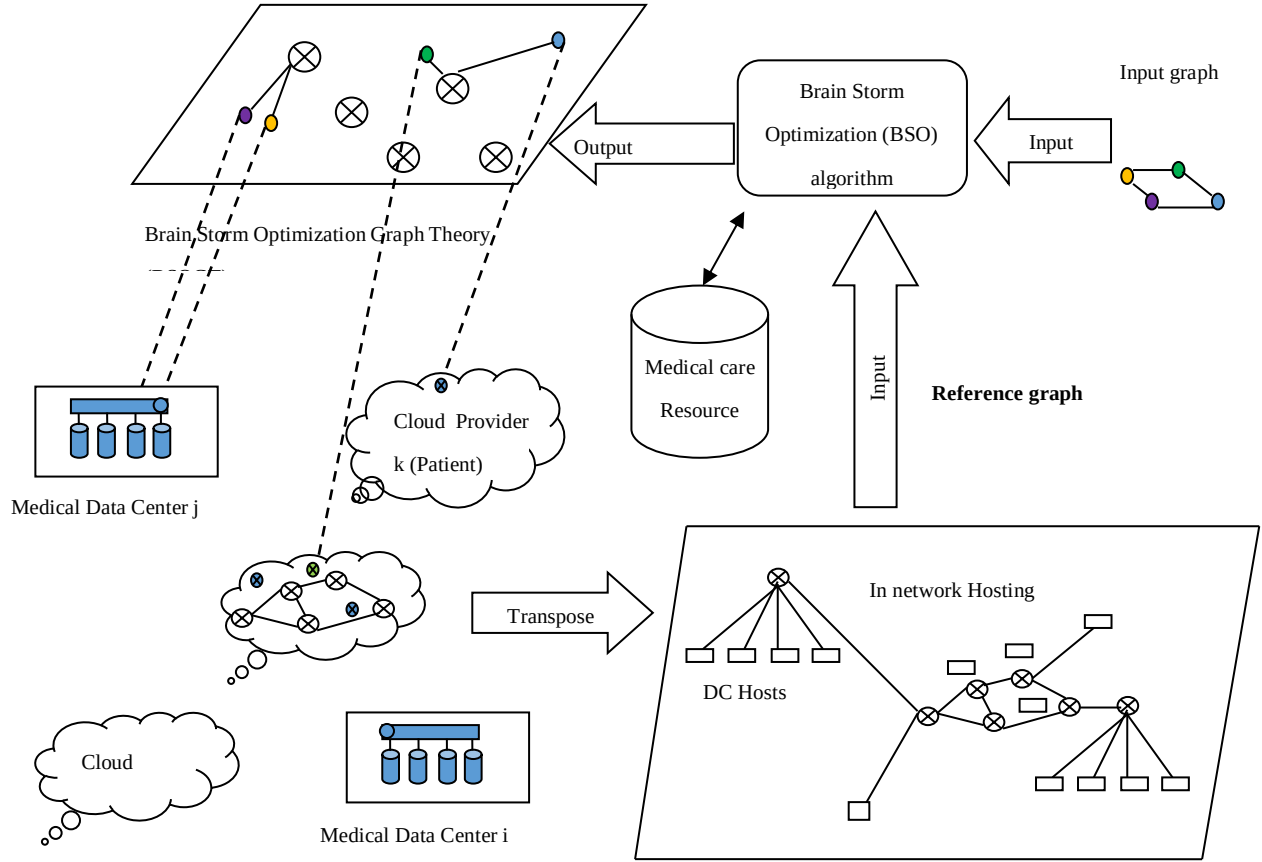


Figure 1: Proposed BSOGT Model

In the case of BSO, a new energy value will be produced in order to replace some of the old energy if the BSO algorithm provides poor mapping results between the physical and virtual network mapping. Whether a BSOGT provides poor mapping results or not, it is decided by the length of the Hamiltonian cycle that can be got by the transfer of every constraint of the physical nodes into an undirected weighted graph. The virtual nodes are mapped onto substrate nodes in a manner that enables the mapping of virtual links onto physical paths in the subsequent phase. The newly introduced algorithm uses graph decomposition into topology patterns by utilizing BSOGT for solving the mapping problem [26]. Rendering both in data center and in-network hosting by considering network nodes to be capable of offering, determining, and storing resources, along with switching and routing functionalities. Figure 1 shows the scope of the work and depicts the inputs and outputs of the newly introduced work (2 data centers, 1 network provider, and 2 public providers, for example). The BSOGT graph is the set of resources from private and public clouds visible to the algorithms. The resources are denoted in this second case by means of a service node bearing attributes and functions provided and are viewed in the form of a container or a hosting platform with the service provisioned with related agreement and quality of service [27]. As shown in Figure 1, the model does the mapping of an input graph comprising nodes and links to the reference infrastructure.

3.1 System Model

The Energy Resource Aware Virtual Network Mapping (ERVNM) problem includes the mapping of a Graph $(G) A = (Ve_A, Ed_A)$ where Ve_A indicates the set of virtual nodes and Ed_A referring to the set of virtual links, to a Brain Storm Optimization Graph (BSOG) $B = (Ve_B, Ee_B)$ in nodes and links (present in vertices and edges). The nodes are ordinary routers, servers, and servers/routers. The set consisting of

available servers in the data centers is denoted as S . The set of routers is indicated by R , and the set of servers/routers is represented by SR . Making use of these notations, the set of vertices of the physical or reference graph B is denoted as $Ve_B = \{S \cup SR \cup R\}$. Therefore, the aim is to search for an injective function $I: Ve_A \rightarrow Ve_B$, which maps every node present in Ve_A to a node in Ve_B , and which matches the edges in Ee_A to edges in Ee_B . That is $\forall(u, v) \in Ee_A(u, v)$ will get matched to $(I(u), I(v))$. This problem is similar to the popular subgraph isomorphism problem explained in [19], which gets an optimal mapping between nodes and edges of two graphs $G1$ and $G2$, given that the number of vertices of $G1$ is less than the number of vertices of $G2$. As illustrated in Figure 1, a tenant requests a mapping of a service comprising 4 interconnected VMs. Depending on the capacity available in the physical infrastructure, every node of the tenant's requested graph will get mapped onto a server or a server/router with the help of algorithms. The equivalences are required for introducing an idea of energy between demand, like the requested virtual resources or CPU, storage, and memory, and chosen physical resources with respect to the input graph and reference graph resources.

$$C(i, j) \Leftrightarrow (c_i \leq C_j) \quad (1)$$

$$ST(i, j) \Leftrightarrow (st_i \leq ST_j) \quad (2)$$

$$M(i, j) \Leftrightarrow (m_i \leq M_j) \quad (3)$$

In this set, c_i (st_i and m_i) is defined as the amount of requested CPU (storage and memory) by VM i , whereas C_j (ST_j and M_j) indicates the available (free) CPU (storage and memory) in a physical node j . Candidate nodes for selection include all of the nodes that satisfy the mapping criteria of equations (1), (2), and (3).

3.2 Brain Storm Optimization Graph Theory (BSOGT)

The important objective of this technical work is to concentrate on the reduction of the embedding cost in addition to the consumption of energy. In the newly introduced BSOG model, the nodes present in the SN will partition into two kinds. In the first active nodes, which take part in the present mapping process n in the form of hosting nodes, are still active from the earlier ones. The energy consumption (E) for the ERVNM process is the energy required for powering the physical resources to the on state, in addition to the energy required for hosting the virtual resources.

$$E_1(i, j, A_{k_1, k_n}) = \begin{cases} \sum_{i=1, \forall j \in m}^n C(ij, k_1, k_n) + ST(ij, k_1, k_n) + M(i, k_n) \\ 0 \text{ otherwise} \end{cases} \quad (4)$$

$$E_2(i, j, A_{k_1, k_n}) = \begin{cases} 1 \text{ if } c_i + c_j \leq C_{k_1} \\ 0 \text{ otherwise} \end{cases} \quad (5)$$

This energy measures how near a physical path is to the actually requested virtual link. These two energies are correspondingly provided by E_1 and E_2 in equations (4) and (5), where $i, j \in Ve_A$ and $k_1, k_n \in Ve_B \setminus R$. A_{k_1, k_n} indicates a path of length n , with k_1 and k_n standing for the endpoints. The above distances provide the expression of the objective function to get the approximate match in nodes and links at the same time for the input graph in the physical graph.

$$\begin{aligned} \min Z = & \min \left[\sum_{i \in Ve_A} \sum_{k \in Ve_B \setminus R} f_{node}(i, k) \times E(i, k) \times x_{ik} + \right. \\ & \sum_{(i, j) \in Ee_A} \sum_{k_1 \in Ve_B \setminus R, k_1 \neq k_n} \sum_{k_1 \in Ve_B \setminus R} f_{link}(ij, k_1, k_n) \times E_1(ij, A_{k_1, k_n}) \times y_{ij, k_1, k_n} + \\ & \left. \sum_{(i, j) \in Ee_A} \sum_{k_1 \in Ve_B \setminus R, k_1 \neq k_n} f_{link}(ij, k_1, k_n) \times E_2(ij, A_{k_1, k_n}) \times y_{ij, k_1, k_1} \right] \end{aligned} \quad (6)$$

where the bivalent variables used are expressed as below

$$x_{ik} = \begin{cases} 1 & \text{if the VM is mapped to } k \in S \cup SR \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

$$y_{ij,k_1,k_n} = \begin{cases} 1 & \text{if the } i \text{ is mapped to } k_1, j \text{ is mapped to } k_n \\ & \text{and } i \text{ is mapped to } A_{k_1,k_n} \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

The first term in the objective function ensures that the demanded resources are available on a candidate physical node k so that it can be chosen. The second term in the objective function gets the optimal physical path A_{k_1,k_n} for supporting the virtual link (i, j) . The third term addresses all of the requests for co-localizing the virtual resources i and j on the same node k_1 . There is a necessity to move to an effective heuristic algorithm to get the convergence times to practical values for implementation in operational networks [20].

In equation (5-6), the energy measure shows how near a physical path is to the actually requested virtual link. The problem of Energy determination is resolved by making use of the BSOGT algorithm. The duty of this BSOGT algorithm is to get a close to optimal solution for ERVNM in a CC environment. In the newly introduced algorithm, all the physical nodes present in the CC model get transferred into an Undirected Weight Graph (UWG) at every mapping of ERVNM. After this, a Hamiltonian cycle and its length are acquired by using a modified cycle algorithm [21]. Then the convergent status of BSO is estimated depending on the ratio of the length of the Hamiltonian cycle at two close generations. When this ratio becomes huge suddenly, some new energy will be created to replace the old ones. This can maximize the diversity of the population and have the BSO escape from the local optima. The weighted graph is represented by $UWG = (V_{UWG}, E_{UWG}, U_{UWG})$, where V_{UWG} stands for a set of nodes, E_{UWG} indicates a set of directed edges, and U_{UWG} stands for a set of undirected edges. The nodes of V_{UWG} represent all of the operations Op_{UWG} , and E_{UWG} is associated with the precedence constraints between the physical and virtual nodes of the parts. U_{UWG} also denotes the set of edges that connect all possible combinations of the nodes.

The process of brainstorming has been applied successfully to solving the ERVNM problem. Intuitively, a BSOGT is developed dependent on the process of human beings' idea generation. The new BSOGT algorithm influenced by the brainstorming process is provided in Algorithm 1 [22-23]. In the process of the BSOGT algorithm depicted in Algorithm 1, Step 1 becomes the initialization step, like in other population-based algorithms; The choice of the initial node decides which requirements of the virtual resources can be met first, which impacts the outcome of ERVNM and the outcomes of BSOGT. Additionally, the undirected edges are combined from the initial physical nodes to the probably first visited virtual machine nodes. After this, steps 2-4 use the process of the physical node 'i' selecting the virtual node j depending on the energy objective function. Step 5a is carried out depending on the randomly chosen physical node i to any randomly chosen virtual link 'j' when its energy objective value is smaller than the probability value, then the current physical node position is changed by choosing the center point. From this, the new energy value is decided at step 5b. when these criteria are not met, then move to else step 6 in algorithm 1, which does the simulation of the new computation of energy value for generating multiple ideas by utilizing the rules; the Step 6a does the simulation of generation of new energy computation influenced by two available concepts from two diverse idea clusters, correspondingly; surely a new energy optimized value can also be an inspiration from more than two available optimized value, but it does not get simulated in the BSOGT for simplifying the algorithm; the number of clusters in Algorithm 1; the cluster center in every cluster acts the objective of better energy optimized value chosen up by ERVNM. Step 6b has the aim of maintaining the generation of better energy-optimized value. The physical node size n determines the simulation of the number of energy optimized values generated in every round of generation of energy optimized values during the process of brainstorming. For simplifying the algorithm, the size of the physical node is fixed to be a constant for every iteration in the BSOGT algorithm.

Algorithm 1: BSOGT algorithm

Begin

1. Randomly form 'O' potential solutions.

2. Cluster (Map) i physical nodes into ' j ' virtual machine node
3. Assess the i physical nodes
4. Rank i physical nodes in every map and record the best individual physical node as cluster center in every map.
5. Randomly generate a new energy value between 0 and 1
 - a) If the value is lesser than a probability pr_a
 - Randomly choose a new physical node ' i ' as map center
 - Randomly create a physical node to replace the chosen map center.
6. Else randomly choose two physical nodes to create new energy value
 - a) Create a random value
 - If it is less than a pre-defined probability p_b , the two physical nodes are merged and then added with random values to create new optimized energy value
 - Else, two energy values from each chosen map are randomly chosen to be merged and added with random values to generate new energy value
 - b) The newly found energy values are compared with the existing energy having the same physical index, the better one is kept and recorded as the new energy
7. If ' n ' new optimized energy values have been created, go to Step 8, else go to Step 6.
8. Terminate if pre-defined maximum number of iterations has been reached; else go to Step 2.

End

The results of matching the A on the B are associated with the matching of the A Patterns on the B patterns generated by the bipartite graph matching. During this stage, just the root nodes of every A Pattern get mapped precisely to the root nodes of the Patterns of B. For meeting the goal of simultaneous node and link mapping, these mappings of roots on roots have to be verified to comply with the latency demands also. In case the target latency is not satisfied, carry out a check on all virtual nodes that are labeled with root and leaves in the order as:

1) *Mapping as a root with leaves*: first by searching and getting a shortest path between the A root (for which there is no confirmation on the latency compliance once the root-to-root mapping in the bipartite graph step that involves A and B is done) and a leaf in the B Patterns in the reference graph;

2) *Mapping as a leaf with roots*: second by the computation of a shortest path between this root considered as a leaf in the other Patterns with the remaining roots and having just the first root, which meets the latency demands;

3) *Mapping as a leaf with leaves*: finally, by also getting a shortest path between the leaf representation of this root found in the other Patterns, and every other leaf, as steps 1 and 2 have not been successful.

4 Performance Analysis

The algorithms were assessed with the help of a 2:53 GHz Quad Core server along with 24 GB of available RAM. The GT-ITM tool [24] was utilized for generating the input and reference graphs. The algorithms are implemented via the use of a cloud simulator in JAVA environment. In the simulations and assessments, for the sake of simplicity, just computational resources (CPU) are taken into consideration in virtual and physical nodes with no generality loss. Random CPU capacity requests ranging between 1 and 5 units are created for every virtual node, and random latency demands ranging between 1- and 100-time units are obtained for every virtual link. For every scenario, 100 independent runs are seen at an average to generate every performance point. These algorithms are implemented via the use of the JAVA environment.

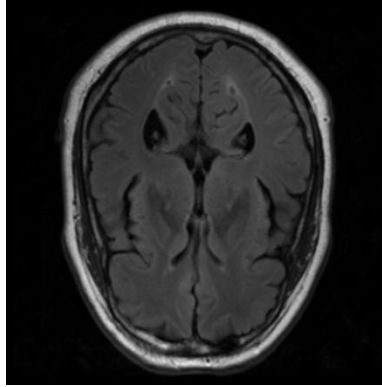


Figure 2(a): Input MRI image of AD- T1 sample 1

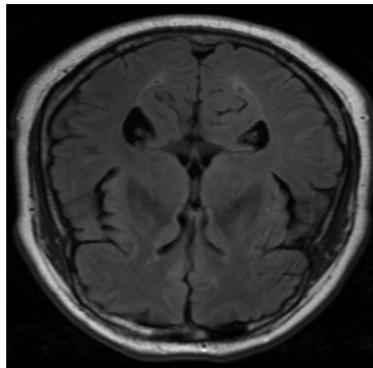


Figure 2(b): Input MRI image of AD- T1 sample 2

The data source consists of hospitals or other sectors that have the Picture Archiving and Communications System (PACS). PACS is an information management scheme that is designed for the management of medical images created by means of radioactive equipment, such as CT, MRE, etc. With the help of the MRI and cognitive data got from the Alzheimer's disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu), the presented method, CORNLIN, was assessed. One objective of ADNI is to test whether serial MRI, positron emission tomography (PET), other biological markers, and medical and neuropsychological evaluation are united to gauge the development of mild cognitive impairment (MCI) and premature AD. For current information, see www.adni-info.org. Figure 2 (a) shows the sample input MRI image of AD- T1 sample 1 and the input MRI image of AD- T1 sample 2.

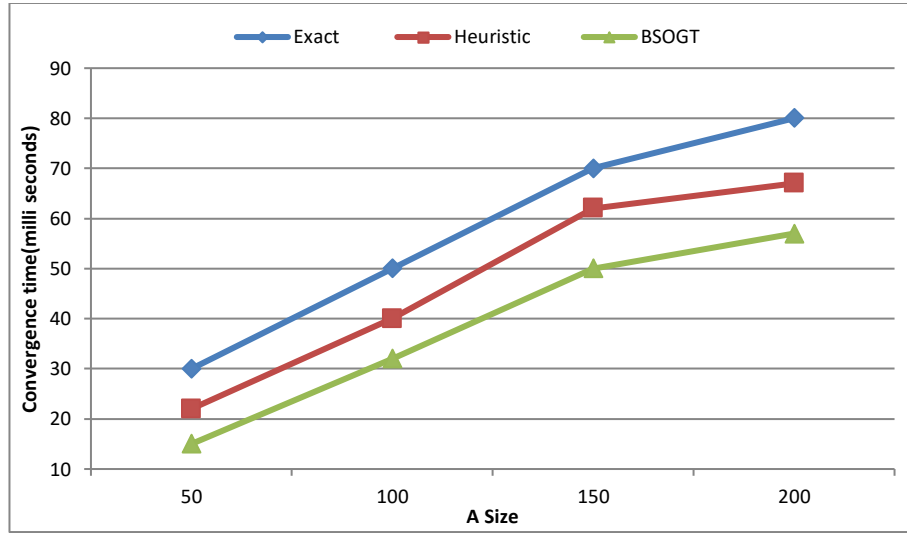


Figure 3: Optimization Algorithms on convergence with B sizes

The first experiment comprises a comparison made between the actual and heuristic algorithms for a B size fixed at 15 virtual nodes. The behaviour in the form of a function of increasing B graph size is found for both algorithms for sensibly small to medium A graph sizes in the range [50, 200]. Figure 3 illustrates the convergence comparison results obtained for the heuristic, exact, and newly introduced BSOGT algorithm. It can be concluded from the figure 2 that the novel BSOGT algorithm outperforms in all the assessed scenarios. The newly introduced work consumes just 57 msec for optimizing the energy values.

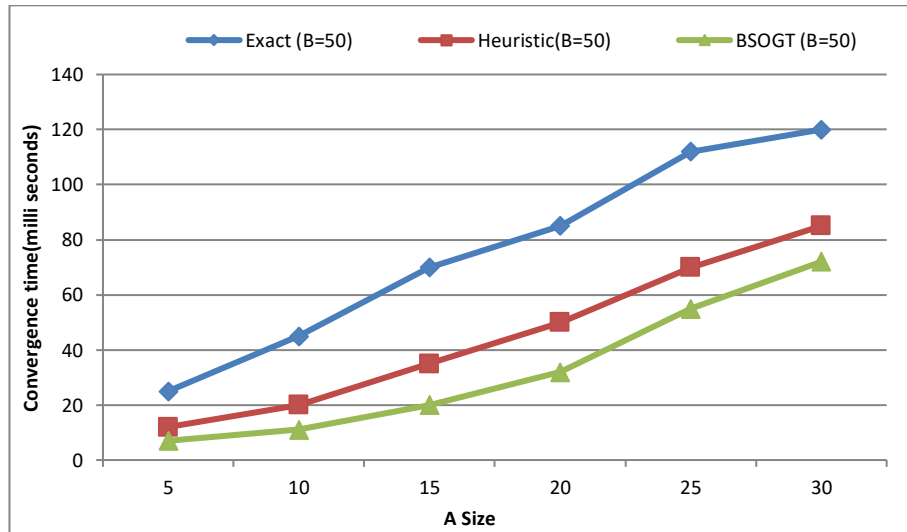


Figure 4: Optimization Algorithms on convergence time with A sizes

As observed in Figure 4, the number of physical nodes is maintained constant at 50 for studying the behaviour of the algorithms with increasing 'A' sizes from 5 to 30 virtual nodes. It is demonstrated from Figure 3 that the newly introduced BSOGT algorithm performs faster for various A sizes, for instance, for A=30 it takes just 72 msec, while other algorithms like exact and heuristic consume 120 msec and 85 msec correspondingly.

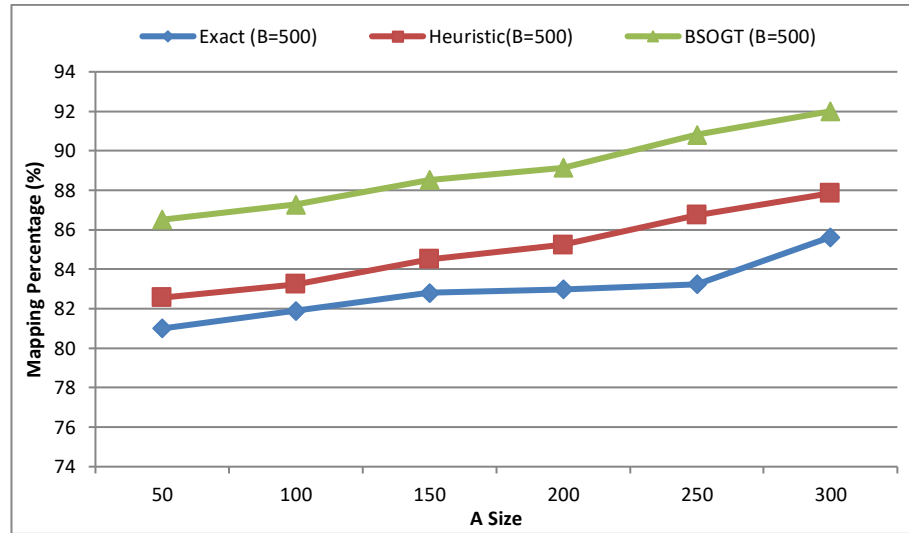


Figure 5: Optimal mapping percentage vs. different Algorithms

For the case of simulated instances existing in the ranges A from 50 to 300 and for B =500 as shown in figure 5, the newly introduced BSOGT algorithm performs very near to optimal and can match the optimal solutions 86.51% to 92% of the runs in nearly not-so-favorable conditions, while the other algorithms like the exact and heuristic yields 85.63% and 87.85% correspondingly. For studying the performance algorithms acceptance ratio, the number of A nodes is taken randomly between 2 and 10. The reference graph is set to 100 nodes.

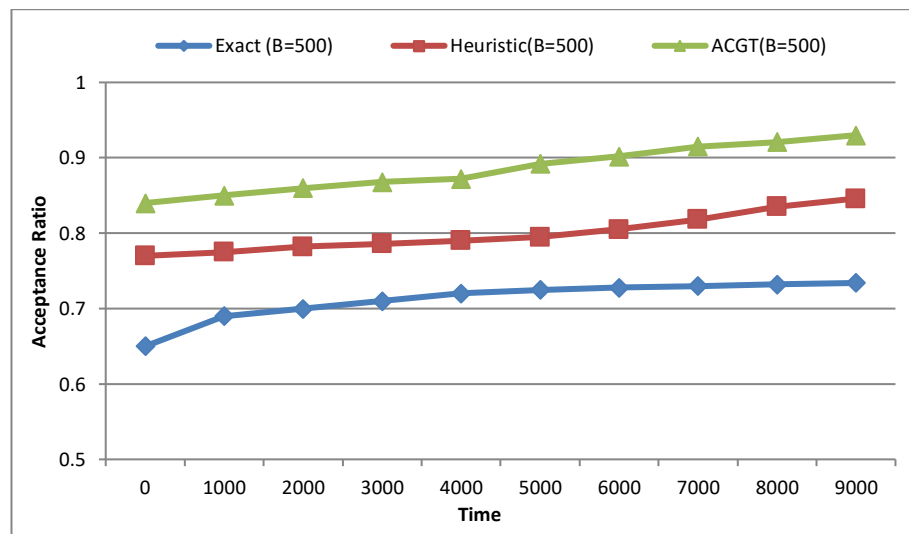


Figure 6: B request acceptance ratio over time

The requested resources (demanded CPU) are obtained and then produced from the Google cluster workload traces given in [25]. The ratio measures the percentage of IG requests that are accepted by an algorithm. The results match with that of those which are found in Figure 7. It can be concluded from Figure 6 that the newly introduced BSOGT algorithm generates 0.93 for a processing time of 9000, while the other algorithms, like the exact and heuristic, yield only 0.734 and 0.846, respectively.

Delays in processing the graphs, particularly the time required for the graph decompositions of the input or reference graph into patterns and the time necessary for updating the reference graph after every mapping, are given in Table 1.

Table 1: ERVNM Processing time comparison vs. different algorithms (A=200 nodes)

A	ERVNM Processing time (ms)		
	Exact	Heuristic	BSOGT
50	42	37	30
100	70	58	52
150	98	90	71
200	130	121	102

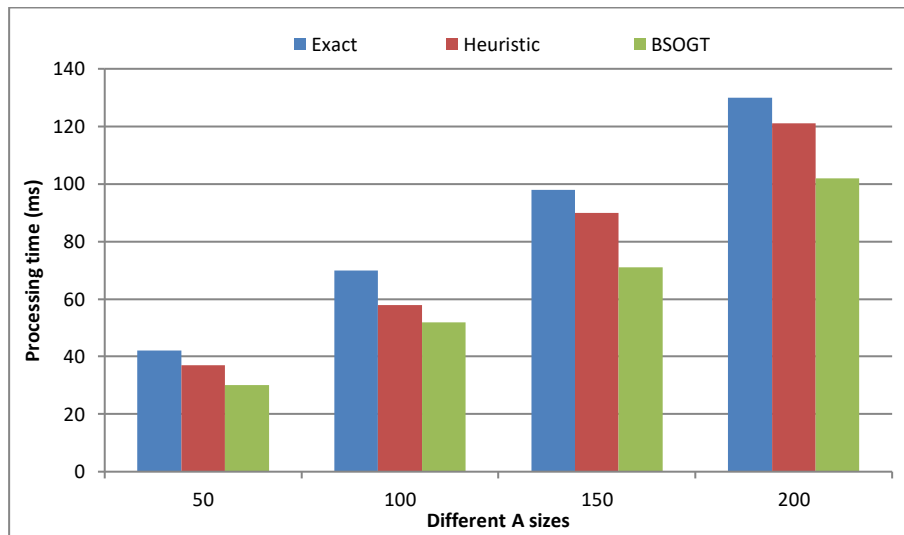


Figure 7: ERVNM Processing Time vs. Optimization Algorithms

Figure 7 illustrates the results of the performance comparison made with the various exact, heuristic, and new BSOGT algorithms, which are measured in terms of the processing time (milliseconds). It can be concluded from the results that the newly introduced BSOGT algorithm incurs very little delay of 102 milliseconds, while the other techniques, like exact and heuristic consumes longer processing times of 130 ms and 121 ms, respectively, for A=200. It is concluded that the new BSOGT algorithm performs better compared to the other two techniques.

5 Conclusion and Future Work

This research work is focused on the significance of Cloud Computing (CC) and network virtualization to satisfy the growth needs for provisioning resources. One of the most essential challenges in resource provisioning is the mapping of the virtual resources to the respective physical ones. This challenge is described as the ERVNM problem. ERVNM is regarded as an NP-hard problem, and it has gained serious attention in the past few years. In this work, a better correlation between the virtual and physical resources is introduced by proposing the Brain Storm Optimization Graph Theory (BSOGT). In BSO, a new energy value will be created for replacing some old energy if the BSO algorithm provides poor mapping results between the physical and virtual network mapping. The objective of BSOGT is about getting a mapping, which balances the load of the hosts about the usage of CPU, with due respect to the limitations enforced by the hosts' resources (memory and storage), and also maintaining the link constraints (bandwidth and latency). The results obtained from the experiments also show that the newly introduced BSOGT algorithm

reduces the time needed to run the experiment on the simulation environment. More work is chiefly focused on the following two aspects: (1) investigating the BSOGT adaptability in various physical network environments and the effect of physical resources on algorithm performance, and (2) exploring the multidomain network issue and trying to minimize the effect of network service outages.

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References

1. Sharkh MA, Jammal M, Shami A, Ouda A. Resource allocation in a network-based cloud computing environment: design challenges. *IEEE Commun Mag.* 2013;51(11):46-52.
2. Fischer A, Botero JF, Beck MT, De Meer H, Hesselbach X. Virtual network embedding: A survey. *IEEE Commun Surv Tutor.* 2013;15(4):1888-906.
3. McAfee A, Brynjolfsson E, Davenport TH, Patil DJ, Barton D. Big data. The management revolution. *Harvard Bus Rev.* 2012;90(10):61-7.
4. Xiong P, Chi Y, Zhu S, Moon HJ, Pu C, Hacigümüş H. Intelligent management of virtualized resources for database systems in cloud environment. In 2011 IEEE 27th International Conference on Data Engineering; 2011 Apr 11, Hannover, Germany; 2011. p. 87-98.
5. Lischka J, Karl H. A virtual network mapping algorithm based on subgraph isomorphism detection. In Proceedings of the 1st ACM workshop on Virtualized infrastructure systems and architectures; 2009 Aug 17, 2009. p. 81-88.
6. Calheiros RN, Buyya R, De Rose CA. A heuristic for mapping virtual machines and links in emulation testbeds. In 2009 International Conference on Parallel Processing; 2009 Sep 22, Vienna, Austria; 2009. p. 518-525.
7. Guo T, Wang N, Moessner K, Tafazolli R. Shared backup network provision for virtual network embedding. In 2011 IEEE international conference on communications (ICC); 2011 Jun 5, Kyoto, Japan; 2011. p. 1-5.
8. Yeow WL, Westphal C, Kozat U. Designing and embedding reliable virtual infrastructures. In Proceedings of the second ACM SIGCOMM workshop on Virtualized infrastructure systems and architectures; 2010 Sep 3, 2010. p. 33-40.
9. Sun G, Yu H, Anand V, Li L, Di H. Optimal provisioning for virtual network request in cloud-based data centers. *Photonic Netw Commun.* 2012;24(2):118-31.
10. Abedifar V, Eshghi M, Mirjalili S, Mirjalili SM. An optimized virtual network mapping using PSO in cloud computing. In 2013 21st Iranian Conference on Electrical Engineering (ICEE); 2013 May 14, Mashhad, Iran; 2013. p. 1-6.
11. Alhazmi K, Sharkh MA, Ban D, Shami A. A map of the clouds: Virtual network mapping in cloud computing data centers. In 2014 IEEE 27th Canadian Conference on Electrical and Computer Engineering (CCECE); 2014 May 4, Toronto, ON, Canada; 2014. p. 1-6.
12. Cao Y, Fan W, Ma S. Virtual network mapping in cloud computing: a graph pattern matching approach. *Comput J.* 2017;60(3):287-307.
13. Alzahrani AS, Shahin AA. Energy-aware virtual network embedding approach for distributed cloud. *arXiv preprint arXiv:1710.11590.* 2017 Oct 31.
14. Xiao X, Zheng X, Zhang Y. A multidomain survivable virtual network mapping algorithm. *Secur Commun Netw.* 2017;2017(1):5258010.
15. Mechtri M, Hadji M, Zeglache D. Exact and heuristic resource mapping algorithms for distributed and hybrid clouds. *IEEE trans cloud comput.* 2015;5(4):681-96.a